

EEAIC assignment2

Alessandro Vegni & Giulio Guerrieri

1 Task

1.a)

The signal gain A_s :

$$A_s = \frac{V_{out}}{V_{in}} = -\frac{Z_f}{Z_{in}} = -\left(\frac{R_f}{1 + sR_fC_f}\right) \cdot (sC_{in}) \quad (1)$$

where:

$$Z_f = R_f \parallel \frac{1}{sC_f} = \frac{R_f}{1 + sR_fC_f} \quad (2)$$

$$Z_{in} = \frac{1}{sC_{in}} \quad (3)$$

Therefore, in the mid-band frequency range (where $sR_fC_f \gg 1$), the signal gain simplifies to the ratio of the capacitors:

$$A_{s,mid} \approx -\frac{C_{in}}{C_f} = -10$$

1.b)

The noise gain A_n is defined as the gain of a fictitious noise signal applied in series with the input terminal of the amplifier. When $A_{OL} \rightarrow \infty$, it is equal to the inverse of the feedback factor β :

$$A_n = \frac{1}{\beta} \quad (4)$$

where β represents the fraction of the output voltage that is fed back to the inverting input of the amplifier when the input source is set to zero. β , in this case, is determined by the impedance divider formed by the input and feedback networks:

$$\beta = \frac{Z_{in}}{Z_{in} + Z_f} \quad (5)$$

Substituting this into the expression for noise gain yields:

$$A_n = \frac{Z_{in} + Z_f}{Z_{in}} = 1 + \frac{Z_f}{Z_{in}} \quad (6)$$

As shown, the noise gain always follows the form of a non-inverting gain formula. By substituting the specific complex impedances, we obtain:

$$A_n = 1 + \left(\frac{R_f}{1 + sR_fC_f} \right) \cdot (sC_{in}) \quad (7)$$

Therefore, in the mid-band frequency range the noise gain simplifies to:

$$A_{s,mid} \approx 1 + \frac{C_{in}}{C_f} = 11$$

It is critical to note that the stability of the amplifier in a closed-loop circuit is determined by the noise gain A_n , not the signal gain A_s . The stability is analyzed via the loop gain $G_{loop}(s) = A_{OL}(s)\beta(s)$. Consequently, the phase margin and system stability are defined at the frequency of intersection between the open-loop gain A_{OL} and the noise gain A_n on a Bode plot.

1.c)

The noise gain A_n represents the actual gain the amplifier provides to any signal appearing at its non-inverting input; therefore, the total output noise is physically the result of the Input Referred Noise open-loop IRN_{OL} multiplied by this A_n .

By taking the total noise observed at the output and mathematically reflecting it back to the signal source via the signal gain A_s one is able to know how much noise is competing with our actual input signal and thus to compute the Input Referred Noise closed-loop IRN_{CL} :

$$IRN_{CL}(s) = IRN_{OL}(s) \cdot \left| \frac{A_n(s)}{A_s(s)} \right|$$

The modulus is used as a negative IRN would imply that adding noise to your circuit somehow increases the precision of your signal, which obviously is impossible.

$$\begin{aligned} IRN_{CL}(s) &= IRN_{OL} \cdot \left| \frac{1 + sC_fR_f + sC_{in}R_f}{1 + sC_fR_f} \right| \cdot \left| \frac{1 + sC_fR_f}{sC_{in}R_f} \right| \\ &= IRN_{OL} \cdot \left| \frac{1 + sC_fR_f + sC_{in}R_f}{sC_{in}R_f} \right| \end{aligned}$$

$$\begin{aligned} IRN_{CL}(j\omega) &= IRN_{OL} \cdot \left| \frac{1 + j\omega R_f(C_f + C_{in})}{j\omega C_{in}R_f} \right| \\ &= IRN_{OL} \cdot \sqrt{\frac{1 + \omega^2 R_f^2 (C_f + C_{in})^2}{\omega^2 C_{in}^2 R_f^2}} \\ IRN_{CL,mid} &\approx IRN_{OL} \cdot \frac{C_f + C_{in}}{C_{in}} = 5.5 \mu V_{RMS} \end{aligned}$$

1.d)

From the definition of Noise Efficiency Factor:

$$NEF := V_{rms,in} \sqrt{\frac{2 \cdot I_{tot}}{\pi \cdot V_T \cdot 4kT \cdot f_{BW}}}$$

Assuming the current consumption is $2\mu A$ and the signal bandwidth is 10 kHz.

- Open-Loop: $NEF = IRN_{OL} \sqrt{\frac{2 \cdot I}{\pi \cdot V_T \cdot 4kT \cdot f_{BW}}} = 2.7289$
- Closed-Loop: $NEF = IRN_{CL} \sqrt{\frac{2 \cdot I}{\pi \cdot V_T \cdot 4kT \cdot f_{BW}}} = 3.0018$

Effect of the input transistor: introduction of a parasitic gate capacitance C_{par} has no impact on the signal gain as C_{par} is connected to a virtual ground, so no signal current flows through it. Conversely, the noise gain increases as C_{par} reduces the feedback factor to

$$\beta = \frac{C_f}{C_f + C_{in} + C_{par}} \quad (8)$$

Therefore, even if the open-loop amplifier remains the same, the closed-loop version becomes noisier and thus any increase in parasitic capacitance directly elevates the IRN_{CL} . Because NEF is linearly proportional to the input-referred noise, C_{par} causes the NEF to rise (less power-efficient design).

2 Task

Cascode configurations have multiple positive advantages (e.g. increasing output impedance and the bandwidth of a circuit without affecting its gain), which is the reason why this topology is used in high-gain amplifier.

The beneficial effects can be extended to three or more stacked devices to achieve higher performances; however, additional voltage headroom is required. Indeed, the minimum voltage overhead that a transistor can occupy is its overdrive.

Note that in the *inverter-based amplifier* configuration the Sources of the cascoded devices ($M_{3a,b}$ and $M_{4a,b}$) are the Drains of the differential inputs ($M_{1a,b}$ and $M_{2a,b}$).

The intermediate node voltage V_{D1} is constrained by the conduction requirement of M_3 , where $V_{GS3} = V_{G3} - V_{S3} = V_B - V_{D1} \geq V_{T3}$, and the saturation condition of M_1 , defined by $V_{GD1} = V_{G1} - V_{D1} = V_{DD}/2 - V_{D1} \leq V_{T1}$ (assumption: input pairs bias at the mid-rail voltage of $V_{DD}/2 = V_B = 0.6V$). These conditions define a permissible range for the drain voltage expressed as $V_{DD}/2 - V_{T1} \leq V_{D1} \leq V_{DD}/2 - V_{T3}$. To ensure a functional voltage window, the threshold of the input pair must exceed that of the cascode devices, such that $V_{T1} > V_{T3}$.

Utilizing high threshold devices (*hvt*) for M_1 and M_2 pairs, while employing low threshold devices (*lvt*) for M_3 and M_4 pairs maximizes the operational margin and transconductance efficiency.

Additionally, this configuration allows the input transistors to operate deeply in subthreshold, maximizing the g_m/I_D ratio to achieve the power and noise limitations. The higher threshold of the input pair effectively lowers the minimum drain voltage required for saturation, while the lower threshold of the cascode ensures the intermediate node V_{D1} can sit at a higher voltage (proof: $V_{GS3} = V_B - V_{D1} = V_{T3} + V_{OV3} \rightarrow V_{D1} = V_B - V_{T3} - V_{OV3}$) and thus providing more headroom for the input transistor pair, ensuring it does not enter the triode region.

3 Task

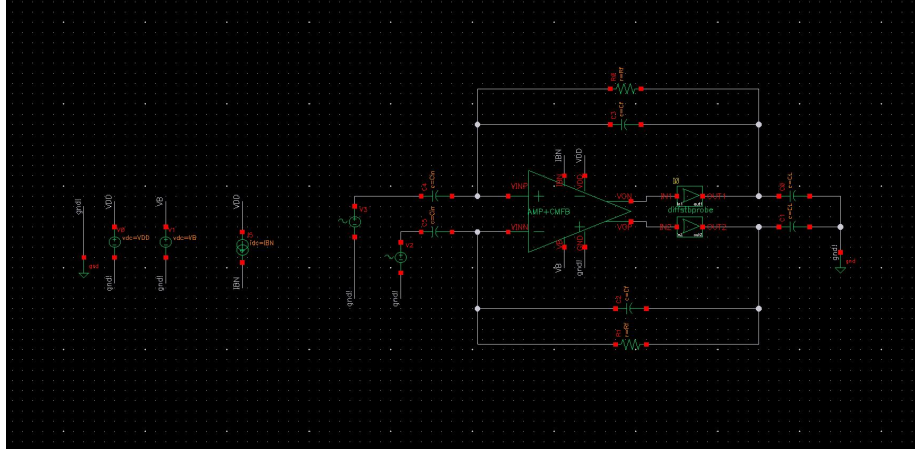


Figure 1: System testbench used for simulation

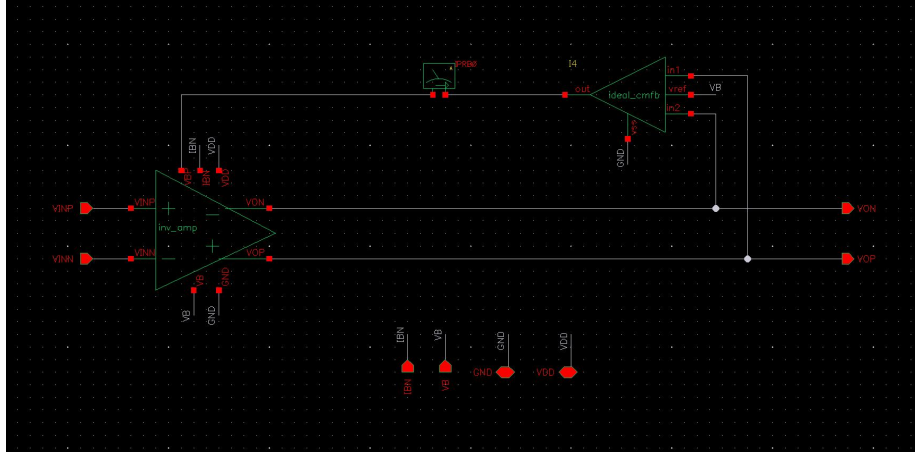


Figure 2: Common mode feedback loop schematic

3.1 Closed loop gain requirement $A_{CL} = 20 \text{ dB}$

We know that $A_{CL} \approx -\frac{C_{in}}{C_f}$ in bandwidth and $C_{in} = 5 \text{ pF}$. From this we initially set $C_f = 0.5 \text{ pF}$ to meet the specific. After the first simulation, since we got a slightly smaller gain than desired we further reduced $C_f = 0.45 \text{ pF}$, moving the first pole ($f_{HP} = \frac{1}{2\pi R_f C_f}$) further by just $\approx 10 \text{ Hz}$.

3.2 Bandwidth High-Pass corner

This is set by the feedback network. The pole is at $f_{HP} = \frac{1}{2\pi R_F C_F}$. We are given $R_F = 10 \text{ G}\Omega$. With $C_F = 0.45 \text{ pF}$, $f_{HP} \approx 39.8 \text{ Hz}$. This satisfies the requirement to pass signals down to 100 Hz.

3.3 Bandwidth Low-Pass corner and transistor sizing

This is dominated by the load capacitance $C_L = 5 \text{ pF}$ and the total effective transconductance $G_m = g_{m,1} + g_{m,2}$. From the conservation of GBWP we know that the open-loop unity-gain frequency $f_{u,OL} = \frac{G_m}{2\pi C_L}$ is equal to the closed-loop bandwidth (f_{LP}) multiplied by the noise gain ($\frac{1}{\beta}$). Thus $f_{LP} = \frac{\beta G_m}{2\pi C_L}$. We will size our transistor widths to ensure that G_m is at least $1,42 \mu\text{S}$ to push this pole past 15 kHz.

3.4 Power Consumption < 500nW

To mitigate potential flicker noise, we initially set the length of the differential pair transistors (M_1 and M_2) to 330 nm , ensuring a sufficiently large device area. Moving to the power budget, with V_{DD} fixed at 1.2 V , the maximum total current allowed for the amplifier is $I_{tot} \approx \frac{500 \text{ nW}}{1.2 \text{ V}} = 416.7 \text{ nA}$. To maximize the transconductance efficiency within this limit, we sized the input transistors to operate in deep weak inversion ($g_m/I_D \approx 28$). By establishing a bias current of 207.8 nA per branch, this setup yields a transconductance $g_m \approx 5.8 \mu\text{S}$. This value comfortably exceeds the bandwidth requirements and sufficiently minimizes the thermal noise contribution to meet the target specifications (see V_{rms} and other parameters of the design in "Figure 6" in the *Scalar Outputs* table).

3.5 Other transistors

The cascode transistor pairs M_3 and M_4 do not determine the bias current themselves but are designed to remain in the saturation region with a small overdrive voltage, ensuring they can carry the current of each branch without restricting the output signal swing.

The head transistor serves as the current source for the structure, with its gate connected to the V_{BP} node. This node is driven by the common-mode feedback circuit, which adjusts the current to keep the output common-mode voltage stable at 0.6 V .

The tail transistor is sized to collect the current from the two differential branches, and its dimensions are chosen to accommodate the total current flow.

3.6 Open-Loop Gain

We derive the open loop gain from the Loop gain $A_{OL} = \frac{A_{loop}}{\beta}$. In mid-band at 1 kHz with $\beta = \frac{1}{13.5}$, $A_{OL, \text{dB}20} = 25 \text{ dB} - 20 \log(\beta) = 47.6 \text{ dB}$

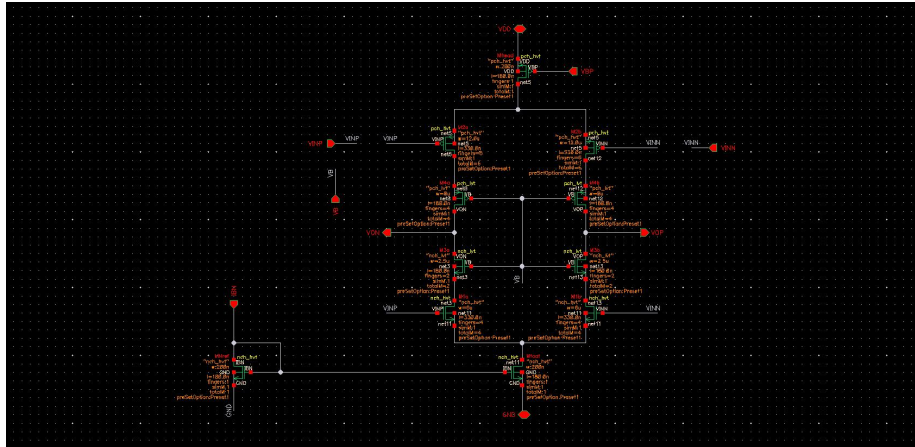


Figure 3: Internal inverter-based amplifier parameters

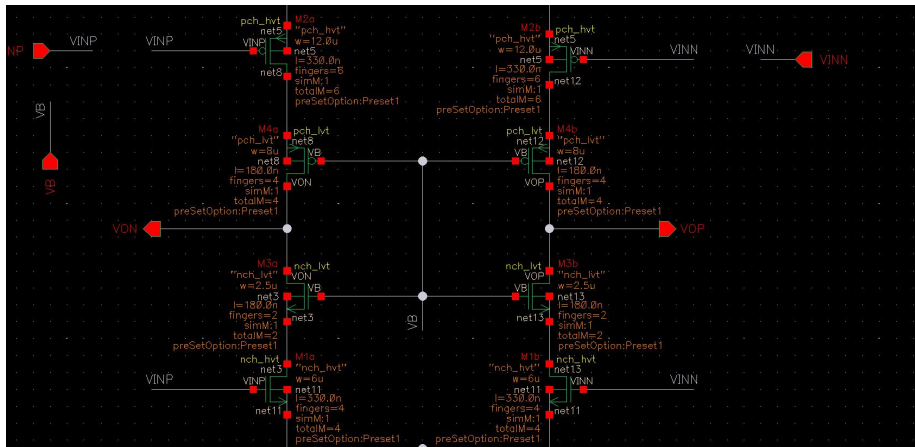


Figure 4: Zoom on parameters

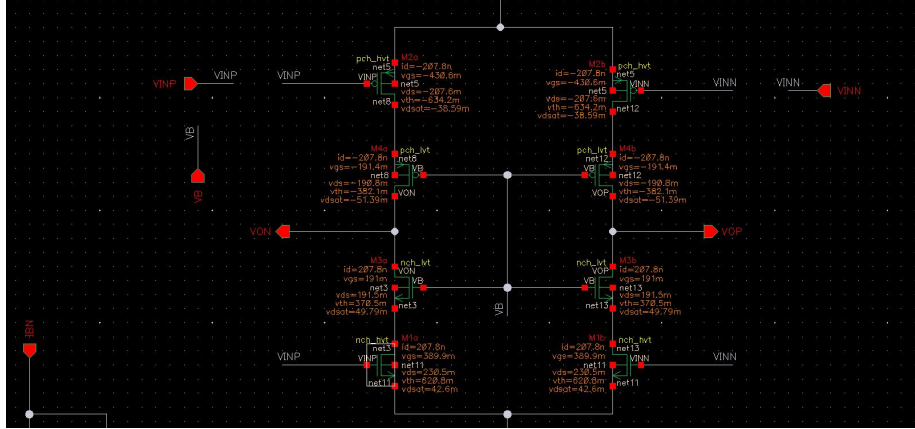
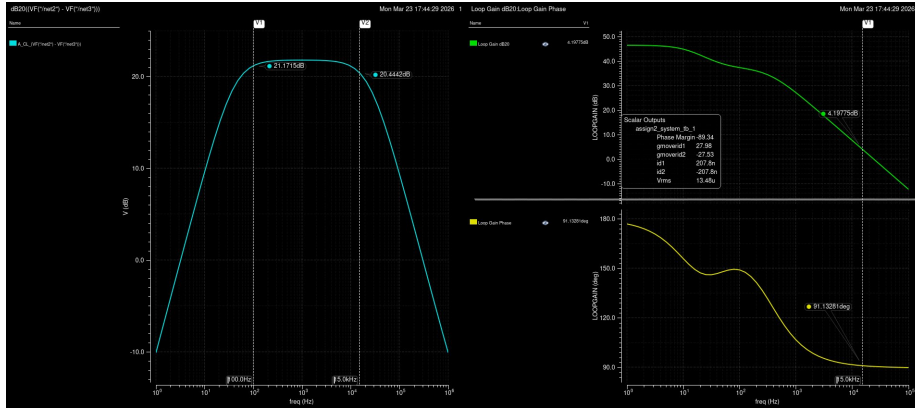


Figure 5: DC Operating Points of the symmetric inverter-based amplifier

Figure 6: Closed/Open-Loop Gain, with Phase diagram, Phase Margin, V_{rms} , transconductance efficiency and branch currents

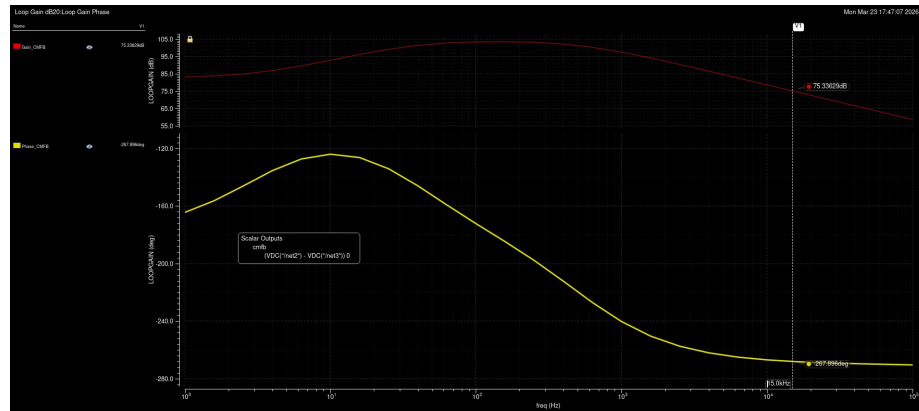


Figure 7: Bode diagrams of Common Mode feedback loop

4 Task

We expect a width offset of the nMOS differential pair to cause a mismatch in the currents flowing in the differential pair branches (as explicit in "Figure 8") and thus a DC voltage offset at the output. The resistive feedback R_f is going to stabilize the circuit, granting a small deviation from ideal operating point.

From simulation we got a DC Offset at the output: $-418.7 \mu V$

Furthermore, differences in the size of nMOS transistor lead to different parasitic capacitances C_{par} at the inputs of the amplifier, causing different noise gain between the two branches. Consequently, common-mode disturbances from the inputs are amplified asymmetrically and therefore these common-mode variations are converted into a differential error at the output, degrading the Common-Mode Rejection Ratio (CMRR).

In a perfectly balanced architecture, any even-order non-linearities produced by either a differential or a common-mode signal appear with identical magnitude and phase in both branches, allowing them to cancel out completely upon differential subtraction. The width mismatch, however, breaks this mathematical cancellation because the second-order distortion terms generated in one branch do not perfectly match those in the other.

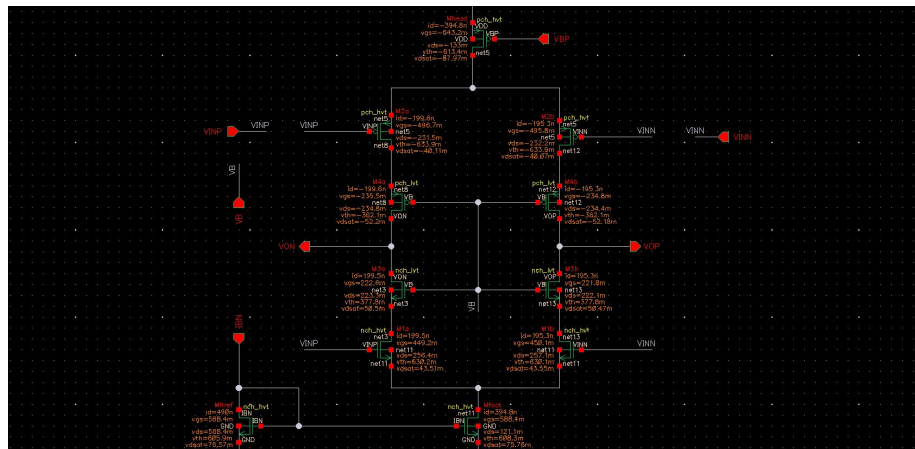


Figure 8: DC operating points with width mismatch

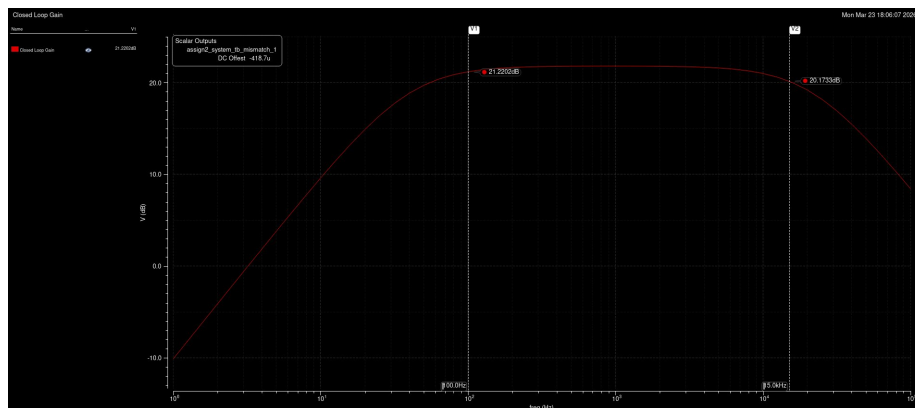


Figure 9: Closed-Loop gain with width mismatch

5 Task

In this design phase, the single *inverter-based amplifier* is expanded into a two-stage chain consisting of *CTA1* and *CTA2*.

Configuration of the First Stage CTA1

To achieve the requested closed-loop gain of $A_{s,1} = 5$ for this specific stage, we decided to keep $C_{in,1} = 5pF$ and set the feedback capacitor $C_{f,1} = 1pF$. This stage remains responsible for most of the signal amplification and noise suppression for the entire chain; indeed, considering the *Friis* formula:

$$v_{ni,tot}^2 = v_{ni,1}^2 + \frac{v_{ni,2}^2}{A_{s,1}^2}$$

Configuration of the Second Stage CTA2

The given load capacitance $C_{L,2} = 100fF$ is very small compared to the $5pF$ $C_{L,1}$ load handled in the single-stage design. Therefore, *CTA2* does not require as much transconductance to maintain a reasonable bandwidth. Furthermore, since the input capacitance of the second stage $C_{in,2}$ acts as the load for *CTA1*, we should choose $C_{in,2}$ so that it does not represent an excessive capacitive load in order to avoid reducing the open-loop gain of the first stage.

To minimize the total Noise Efficiency Factor ($NEF \propto V_{n,rms|tot} \cdot \sqrt{I_{tot}}$), we must look at how noise is distributed across the chain.

The total input-referred noise (*IRN*) is dominated by the first stage because any noise generated by *CTA2* is divided by the gain of the first stage when referred back to the input. Therefore, spending current on the second stage to lower its individual noise floor provides very little benefit to the system performance, this is the reason why *CTA2* is biased with a very low current that anyway is sufficient to meet the speed requirements as it only needs to drive a $100fF$ load.

The majority of the current budget should be allocated to *CTA1*. Increasing current in the first stage directly increases gm and lowers thermal noise, which has a significant impact on the total *IRN* and the NEF.

Energy-efficient Analog Integrated Circuits

Sensor Interface – Exercise 2

Assignment

Gabriele Atzeni

Prof. Taekwang Jang

1. Consider the amplifier depicted in Fig. 1, where $IRN = 5 \mu V_{RMS}$ is the input-referred noise of the open-loop amplifier. Consider $C_{IN} = 10 \text{ pF}$ and $C_F = 1 \text{ pF}$ and $R_F = 10 \text{ G}\Omega$.
 - a. Calculate the signal gain assuming the amplifier open-loop gain $A_{OL} \rightarrow \infty$.
 - b. What is the noise gain A_n ?
 - c. What is the input-referred noise of the closed-loop amplifier?
 - d. Calculate the NEF of the open-loop amplifier and closed-loop amplifier assuming the current consumption is $2 \mu A$ and the signal bandwidth is 10 kHz .
 Discuss the effect of the parasitic input capacitance introduced by the input transistors of the amplifier on noise and NEF.

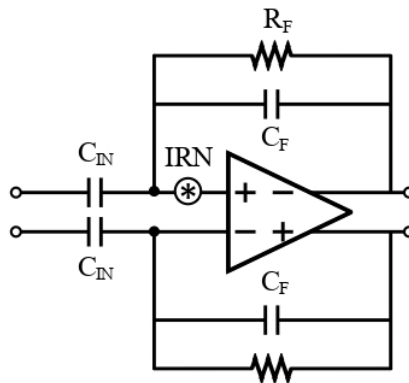


Fig. 1: Amplifier with capacitive feedback

2. Fig. 2 shows the schematic of an inverter-based amplifier with its common-mode feedback circuit. Assume $V_B = V_{DD}/2$ and $V_{DD} = 1.2 \text{ V}$. How would you choose the threshold voltage of M_1 and M_2 compared to M_3 and M_4 so that M_3 and M_4 operate as cascode devices?

3. Design the amplifier to meet the following specifications and explain your design strategy carefully:
 - Open-loop gain $A_{OL} > 40$ dB
 - Closed-loop gain $A_{CL} = 20$ dB
 - Bandwidth: 100 Hz – 15 kHz
 - Load capacitance $C_L = 5$ pF
 - Phase Margin $PM > 60^\circ$
 - Common-mode feedback phase margin $> 60^\circ$
 - Power consumption < 500 nW
 - Input-referred noise $IRN < 20 \mu V_{RMS}$
 - $NEF < 3$
4. Assume $R_F = 10$ G Ω and $C_{IN} = 5$ pF. Note: you need to use the diffstbprobe component from analogLib to run the stability analysis (stb) of a fully differential amplifier. Refer to Fig. 3 to check the probe configuration for the capacitive feedback and common-mode feedback. An ideal common-mode feedback circuit is available at /home/eeaic_04fs26/ExerciseTools. You

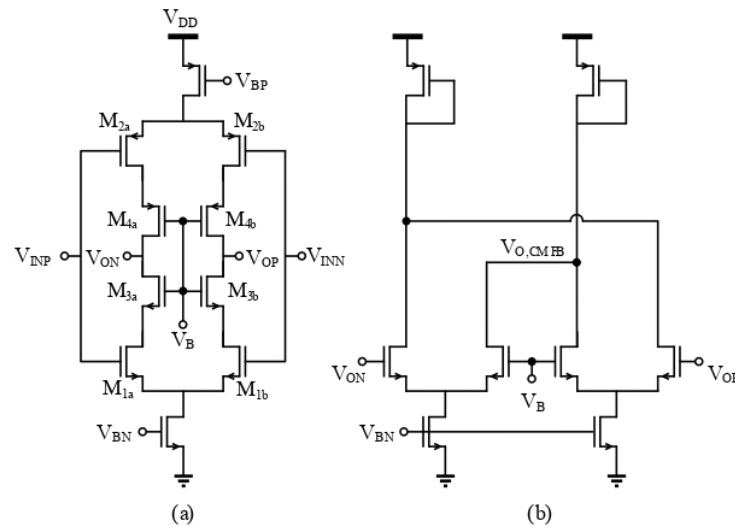


Fig 2: Inverter-based cascode amplifier (a), and its common-mode feedback circuit (b)

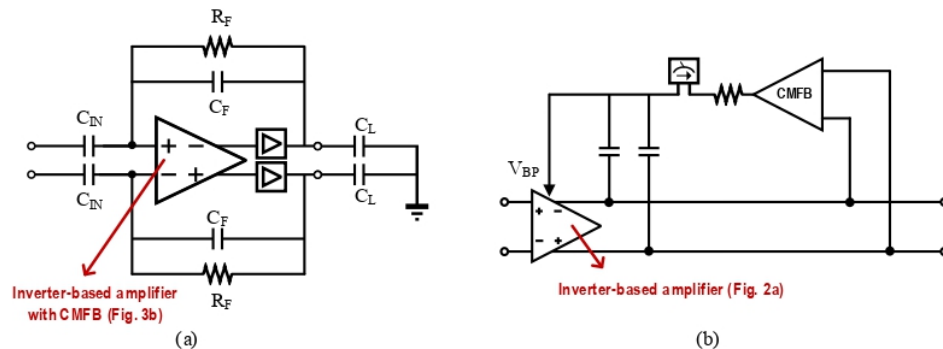


Fig. 3: Probe configuration for stability analysis of the capacitive feedback loop (a) and common-mode feedback loop (b).

can use `cp -r /home/eeaic_04fs26/ExerciseTools/ideal_cmfb <your cockpit folder>/dfi/<your library>` to copy the ideal model in your working library. Alternatively, you can design the common-mode feedback circuit using the topology in Fig. 2a. Discuss qualitatively the effect of a mismatch in the width of M_{1a} and M_{1b} . Verify your answer by introducing a 5% mismatch in simulation.

5. Assume your inverter-based amplifier is used as the first stage (CTA1) of the amplifier chain depicted in Fig. 4. Add a second stage (CTA2) and set C_{IN1} and C_{F1} to achieve a closed-loop gain of 5 for the first stage. Assume the load capacitance of the second stage is 100 fF. How should you set the current consumption of CTA2 to minimize the overall NEF?

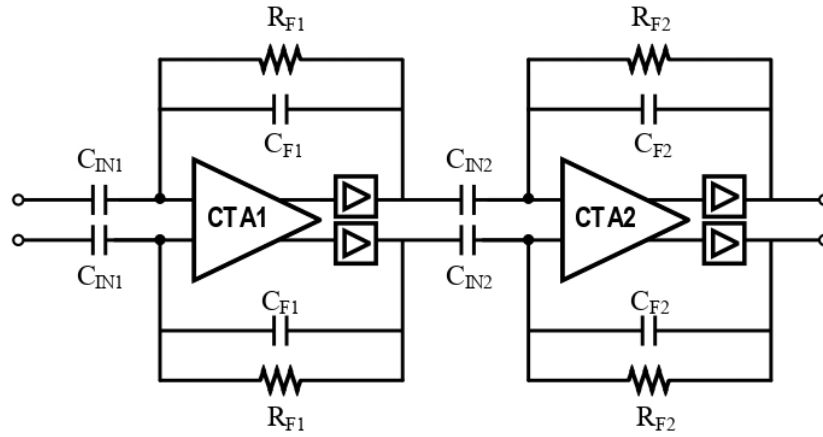


Fig. 4: Amplifier chain consisting of two inverter-based amplifiers with capacitive feedback